NONEMPTY CLOSED SUBSETS OF PROJECTIVE SPACE WHOSE DIMENSIONS SUM TO AT LEAST N INTERSECT, WITH A DIMENSION BOUND FOR THE INTERSECTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites four cross-chapter labels, each recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The author's parenthetical 'details omitted' concerns only the elementary verification that pi is surjective and smooth of relative dimension one with irreducible fibres; the main cone argument is written out.

Allowed standard prerequisites (editorial).

- The affine-space intersection bound. Stacks Project, Varieties, Lemma lemma-intersection-in-affine-space (tag 0A2T). Provides the dimension bound once the cones have been arranged to meet at the origin.
- Dimensions of fibres and of preimages under a smooth morphism. Stacks Project, Varieties, Lemma lemma-dimension-fibres-locally-algebraic (tag 0B2L). Gives the dimension of the fibres of the restrictions of V and W used to show they contain every fibre they meet.

Source reference. Original reference 01NE: sources/stacks-chow/tags.txt, source lines 2155–2155. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0B2L: sources/stacks-chow/tags.txt, source lines 13934–13934. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0B2Q: sources/stacks-chow/tags.txt, source lines 13938–13938. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field and $n \geq 0$. Let $X, Y \subset \mathbf{P}^n_k$ be nonempty closed subsets. If $\dim(X) = r$ and $\dim(Y) = s$ and $r + s \geq n$, then $X \cap Y$ is nonempty and $\dim(X \cap Y) \geq r + s - n$.*

Proof. Write $\mathbf{A}^n = \mathrm{Spec}(k[x_0, \ldots, x_n])$ and $\mathbf{P}^n = \mathrm{Proj}(k[T_0, \ldots, T_n])$. Consider the morphism $\pi : \mathbf{A}^{n+1} \setminus \{0\} \to \mathbf{P}^n$ which sends $(x_0, \ldots, x_n)$ to the point $[x_0 : \ldots : x_n]$. More precisely, it is the morphism associated to the pair $(\mathcal{O}_{\mathbf{A}^{n+1}\setminus\{0\}}, (x_0, \ldots, x_n))$, see Constructions, Lemma 01NE. Over the standard affine open $D_+(T_i)$ we get the morphism associated to the ring map

$$k\left[\frac{T_0}{T_i}, \ldots, \frac{T_n}{T_i}\right] \longrightarrow k\left[T_0, \ldots, T_n, \frac{1}{T_i}\right] \cong k\left[\frac{T_0}{T_i}, \ldots, \frac{T_n}{T_i}\right]\left[T_i, \frac{1}{T_i}\right]$$

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which is surjective and smooth of relative dimension 1 with irreducible fibres (details omitted). Hence $\pi^{-1}(X)$ and $\pi^{-1}(Y)$ are nonempty closed subsets of dimension $r+1$ and $s+1$. Choose an irreducible component $V \subset \pi^{-1}(X)$ of dimension $r+1$ and an irreducible component $W \subset \pi^{-1}(Y)$ of dimension $s+1$. Observe that this implies V and W contain every fibre of π they meet (since π has irreducible fibres of dimension 1 and since Lemma 0B2L says the fibres of $V \to \pi(V)$ and $W \to \pi(W)$ have dimension ≥ 1). Let $\overline{V}$ and $\overline{W}$ be the closure of V and W in $\mathbf{A}^{n+1}$. Since $0 \in \mathbf{A}^{n+1}$ is in the closure of every fibre of π we see that $0 \in \overline{V} \cap \overline{W}$. By Lemma 0B2Q we have $\dim(\overline{V} \cap \overline{W}) \geq r+s-n+1$. Arguing as above using Lemma 0B2L again, we conclude that $\pi(V \cap W) \subset X \cap Y$ has dimension at least $r+s-n$ as desired. $\square$